Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/02/2004
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Project Title:	Advanced Optical Frequency Standard for Space		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305-4008	**Congressional District:**	15
Comments:	NOTE: Previously at National Institute of Standards & Technology (Ed., Fall 2011)		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth"s gravity		

To develop high-accuracy high-stability optical frequency standards for space based on laser-cooled and trapped neutral atoms and ions. Microwave frequency standards have achieved higher stability and accuracy in the last decade due in large part to laser cooling and trapping [1,2,3]. Better stability and accuracy will be difficult since microwave clocks are now near their fundamental performance limit. However, if the frequency of the standard could be made higher, then its stability (and possibly its accuracy) could be proportionally higher [4]. Thus, optical oscillators that are referenced to narrow resonances in laser-cooled atoms will be the time and frequency standards of the future. Our research focuses on eliminating or reducing the most serious impediments to the performance of optical frequency standards.
These include: 1.Acceleration-induced noise in optical reference cavities 2.Gravitation limitation to the interaction time for neutral-atom clocks 3.An optical frequency synthesis/measurement system that preserves the phase fidelity of the optical references

Task Description:	We are evaluating the performance of two optical frequency standards, one based on a trapped and laser-cooled single Hg+ ion and the other on laser-cooled calcium atoms. Since the fractional frequency instability of atomic clocks is inversely proportional to the frequency of the clock transition and to the square root of the Ramsey interrogation period, greatly improved performance compared to microwave frequency standards is possible by utilizing a narrow optical transition in laser-cooled atoms. Projected fractional frequency instabilities and inaccuracies for laser-cooled, stored ion optical frequency standards that are earth based are about 1x10-18, more than 2 orders of magnitude better than that projected for atomic microwave frequency standards even in a micro-gravity environment. However, optical frequency standards based on laser-cooled neutral atoms in a micro-gravity environment could potentially produce even higher stabilities. In order to realize their projected stabilities, both types of optical frequency standards require a spectrally pure local oscillator that has a short term (<10 s) instability less than 5x10-16. Most often, this is achieved by locking the frequency of the laser to a high-finesse and well-isolated optical cavity. Frequency fluctuations of the optical oscillator are then dominantly caused by seismic perturbations that are coupled through the suspension system which must support the cavity against the pull of gravity. Since the fundamental resonance of the isolation support system is proportional to the sqrt(g, seismic isolation could be more than 3 orders of magnitude better in a micro-g environment. Finally, the full benefits of a highly stable and accurate optical frequency standard are available only by counting the zero crossings of the frequency of the optical standard to generate time. We plan to thoroughly investigate broadband optical synthesis schemes using pulsed lasers to connect visible frequencies to countable microwave frequencies in a phase coherent manner. Recent progress in our laboratory and others shows that mode-locked fs-lasers coupled to photonic fibers can be used to span large frequency intervals in the visible spectrum and to measure optical frequencies with high precision and accuracy. The many orders of magnitude improvement expected in the performance of frequency standards based on optical transitions will open many new opportunities in experimental fundamental physics, including: improved tests of relativity, searches for time variation of fundamental constants, precision time and frequency metrology. We can also anticipate applications in advanced communication systems and navigation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have made good progress in FY04 on our NASA funded project for optical frequency standards for space. This includes several significant improvements in the single, 199Hg+ optical frequency standard. In the past year, we have added a second, polarization-modulated 194 nm beam, which permits us to operate the standard near zero field without optical pumping into dark states and loss of fluorescence. Hence, we have been able to make absolute frequency measurements with a small 2nd order field shift (typically 20 Hz vs. 3 kHz previously) and a correspondingly smaller frequency uncertainty caused by any field fluctuations and drift. Most recently, by comparison to the Ca optical frequency standard and to our well-isolated reference cavity, we have made a measurement of the 199Hg+ quadrupole moment (roughly = -1.86 x 10^-21 em^2). This is smaller than originally anticipated and gives rise to a maximum frequency shift of about 0.5 Hz/V on a single endcap. Typically, our contact term is no more than 0.1 V. This corresponds then to a maximum frequency shift of 50 milliHz or a fractional shift of 5 x 10^-17. We have also made important strides toward the realization of the two ion, quantum-entangled optical clock, which uses Be+ as the logic ion and Al+ as the clock ion. In the past 6 months, we have been able to load and sympathetically cool Al+, by Doppler cooling the Be+ ion. In fact, all motional degrees of freedom have been cooled to the n=0 ground state of motion. In the past month the optical clock transition in Al+ has been detected via the transfer of its state information to the Be+ ion via the motional coupling of the pair. Soon, we expect to operate this system as a clock and measure its frequency against the NIST Cs primary standard. For the optical clock based on laser-cooled neutral Ca atoms, with the dominant systematic shifts due to Doppler effects now virtually removed from the apparatus, we have started a detailed investigation of other systematic shifts. We have developed an interlaced measurement scheme that takes advantage of the standard's exceptional short term stability that allows us to test possible frequency shift mechanisms. With only minutes of averaging time we can now evaluate potential shifts at the Hz level. We anticipate a new measurement of the Ca frequency in the upcoming months with a fractional frequency uncertainty well below that of other existing neutral atom standards. We have also expanded our neutral atom optical clock program with the construction of a second optical clock, based on laser-cooled Yb, which has enormous potential for high accuracy and stability. We recently demonstrated an Yb magneto-optic trap that uses laser diodes at 399 nm and traps more than one million atoms. We plan to load the Yb atoms into a specially designed optical lattice, which will greatly reduce systematic shifts for the clock transition at 578 nm, which has a natural linewidth of only 15 mHz.
Bibliography Type:	Description: (Last Updated: 09/03/2025)
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